

Nitish Agrawal

Design Verification Engineer | M.Tech, IIT Delhi

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WORK EXPERIENCE

Design Verification Engineer | Qualcomm India Pvt Ltd., Bangalore

July 2024 - Nov 2025

► Camera Verification

- Extracted functional requirements from micro-architecture documents to define block-level specifications and test plans; collaborated with the design team to ensure design correctness.
- Identified and root-caused a power-sequencing RTL bug using force statements to bypass the faulty signal and confirm downstream path integrity
- Identified an SoC-level register accessibility bug caused by an SMMU stream-ID mismatch by correlating SMMU fault logs with register access failures—both fixes were confirmed and corrected in the design.
- Verified camera subsystem functionality across 5 clock speed configurations and multiple power collapse and re-initialization scenarios, validating register state retention, clock/reset sequencing, and error-free frame transfer resumption post-collapse.
- Provided regular verification status updates, collaborated across design, validation, and methodology teams.

► LLM-Based Automation for Test Sequence Generation

- Designed and deployed an LLM-driven model to generate C and SystemVerilog test sequences for multimedia and security IPs, with targeted prompt engineering for correct register programming and power & clock sequencing.
- Led prompt engineering for fine tuning the model for correct register programming and power & clock sequencing.
- Reduced manual effort by ~30–40%, allowing verification engineers to focus more on verification strategy rather than boilerplate code.

Research Intern | Cadence Design Systems, Noida

July 2023 - May 2024

► Thermal-Driven PCB Macro Placement Optimization

- Modeled components using a 2-resistor thermal network derived from four key temperature points and extended to multi-component systems via superposition; analyzed copper routing effects on thermal conduction.
- The optimized placement strategy reduced local temperature peaks and provided a predictive macro-positioning framework based on power profiles, improving design efficiency and thermal performance.

TECHNICAL SKILLS

Methodology:	Testplan Development, UVM, Constrained Random Verification, SV Assertions, Functional Coverage
Protocols:	AXI4, APB, MIPI
Languages & Scripting:	System Verilog, Python
EDA Tools:	Synopsys VCS, Synopsys Verdi, Cadence Virtuoso, Innovus, Tempus, IVerilog, GTKWave

EDUCATION

Masters in VLSI Design Tools and Technology	IIT Delhi	7.6/10	Aug 2022 - May 2024
B.E. in Instrumentation and Electronics	Jadavpur University	8.0/10	Aug 2018 - May 2022

PROJECTS

Synchronous FIFO Verification | GitHub Repository

Mar 2026 - April 2026

- Developed a SystemVerilog-based UVM-style layered testbench comprising transaction, driver, and monitor classes.
- Verified functional coverage of FIFO boundary conditions with directed testcases (reset, overflow, underflow, data integrity) and a self-checking scoreboard achieving 12/12 pass count.

AXI4 UVM Verification | GitHub Repository

Feb 2026 - Mar 2026

- Designed a synthesizable AXI4-Lite DUT in SystemVerilog with separate write/read channel FSMs and SLVERR error response.
- Built a UVM testbench hierarchy with sequencer-driver handshake, dual TLM analysis ports, and a functional coverage model capturing burst types, transfer sizes, boundary conditions, and FSM cross coverage.

128k * 8 SRAM Memory Design

Prof. Kaushik Saha | Jan 2024 - April 2024

- Designed memory array architecture and sense amplifier circuits in a 4-member team using Cadence Virtuoso & extracted parasitics (300–350 fF bitline, 40–60 fF wordline).
- Achieved read access time of 5–5.5 ns and write time of 6–7 ns; latch-type sense amplifier with 70–90 mV minimum differential input and 350–450 ps sensing delay.
- Designed a latch-type sense amplifier with 70–90 mV minimum differential input and 350–450 ps sensing delay.

8-bit 16-depth Synchronous FIFO

Prof. Jayadeva | Jan 2023 - April 2023

- Designed a synchronous FIFO in Verilog using an extra-bit pointer scheme to distinguish full from empty; developed control logic for simultaneous read-write with boundary condition integrity.
- Executed a complete RTL-to-GDSII flow on Cadence Genus and Innovus, resolving setup/hold violations for timing closure with final signoff in Cadence Tempus.

Mod-6 Up/Down Counter

Prof. Jayadeva | Oct 2022 - Nov 2022

- Designed schematic & layout using Cadence Virtuoso on UMC 65nm
- Used Gray-code sequencing to minimize switching activity; optimized power/ground routing on M6/M5 and clock routing on M4.
- Achieved 3.2 GHz operation, 32 μ W power, and 113 μ m² layout area with DRC & LVS clean checks.

Branch Predictor Design

Prof. S.R. Sarangi | Sept 2022 - Nov 2022

- Pag predictor used GHR-based per-address tables and achieved 98% accuracy on test traces.
- Tournament predictor integrated local & global predictors with a 2-bit chooser and delivered 97% accuracy.